4.2.1 Measuring improvement with a Gaussian process

In this subsection, we examine how improvement in BayesOpt is defined, how it constitutes a good measure of utility, and that working with improvement-related quantities is straightforward with normal distributions.

As our final goal in BayesOpt is to identify the global optimum of the objective function—the point that gives the highest objective value—the higher the values we observe while making evaluations of the objective function, the higher our utility should be. Suppose as we make an evaluation of the objective function at some point x_1, we observe the value 2. In another situation in which we evaluate another point x_2, we observe the value 10. Intuitively, we should value the second point more, as it gives us a higher function value.

However, if before observing x_1 and x_2, we have already seen a point x_0 that yields a value of 20. In this case, it's natural to think that even though x_2 is better than x_1, neither points result in any additional utility, as we already have a much better observation in x_0. On the other hand, if we were to have a point x_3 that yields a value of 21, then we would be happier, as we would have found a better value than that of x_0. This is illustrated in Figure 4.7.

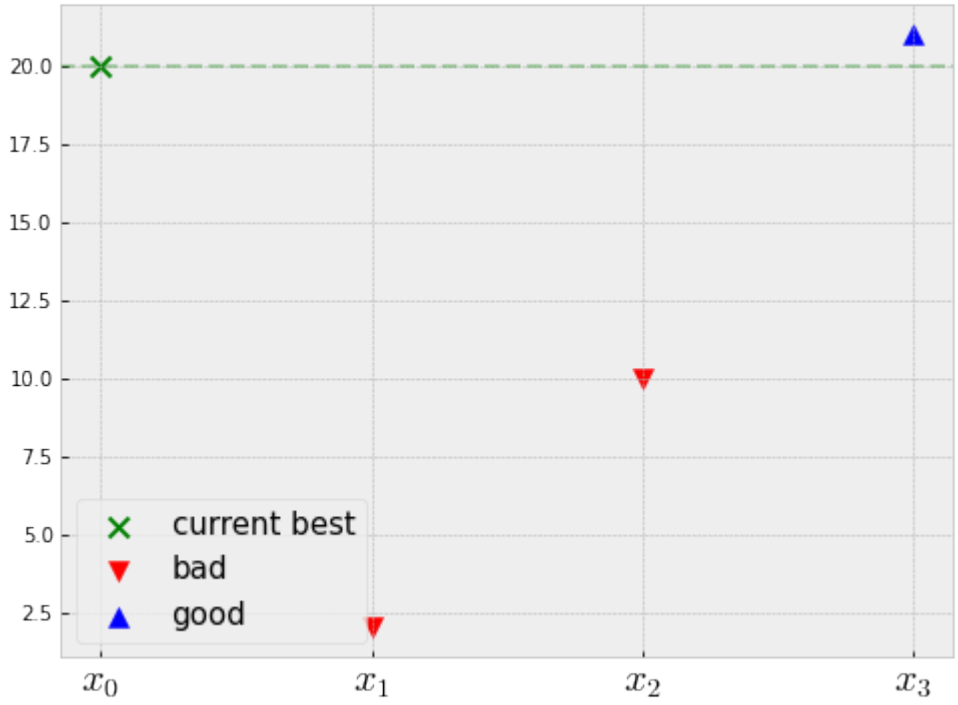


Figure 4.7 Seeking improvement from the best-seen point. Although point x_2 is better than point x_1, both are "bad" in the sense that they don't improve from point x_0.

These comparisons point to the idea that in BayesOpt, we care about not only the raw values of our observations, but also whether the newly found observation are better than the observations

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